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*Article*

# Supplementary Materials: WRF Cold-Pool Sensitivities Relevant to Winter-Ozone Modelling in the Uinta Basin

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**Abstract**

This document collects bounded secondary diagnostics that support the main article. It does not change the controlled driving-product result or the status of any predeclared gate.

These supplementary materials retain bounded secondary diagnostics from the 2 February 2013 Uinta Basin WRF case. They do not change the article's controlled driving-product result or the status of its failed gates.

**Table S1.** Controlled process-treatment effects on the frozen station/common-AGL pipeline. Values are means of five hourly spatial medians; signs are positive/negative hours with exact zeros omitted. Planetary boundary-layer height is scheme defined across the boundary-layer/surface-layer treatment.

| Metric | Slope radiation (X1 − X0) Mean | Slope radiation (X1 − X0) Signs | MYJ/Eta (X2 − X1) Mean | MYJ/Eta (X2 − X1) Signs |
|---|---|---|---|---|
| T2 bias (°C) | +0.001 | 3/1 | +0.37 | 5/0 |
| 10 m wind absolute error (m $s^{-1}$) | +0.001 | 2/2 | -0.145 | 0/5 |
| Surface-pressure bias (Pa) | -0.12 | 0/4 | +6.3 | 5/0 |
| Stability $S_{50:500}$ (K) | +0.004 | 3/1 | +0.11 | 3/2 |
| Transport wind $U_{50:500}$ (m $s^{-1}$) | +0.006 | 4/0 | +0.098 | 5/0 |
| PBL height (m) | +0.01 | 3/1 | -9.2 | 2/3 |

**Table S2.** Static-terrain diagnostics retained after the failed default-minus-fine gate and the closed coarse-member lineage check.

| Contrast | Terrain difference | 2 m response | Status |
|---|---|---|---|
| Default minus fine | 7.9 m RMS; -55 to +60 m range | Elevation-band means 0.07–0.27 K; terrain correlation below 0.3 at 1400–1600 UTC | Predeclared conjunction failed |
| Coarse brackets | 91.9/93.1 m RMS; extrema to -462/+487 m | Slope-band means 1.2–1.5 K; terrain correlation 0.50–0.67 | Descriptive only; no predeclared gate |
| Coarse-member lineage | As-run namelist matches the controlled family; terrain is materially smoother | Configuration caveat closed | Supports controlled-bracket status, not a terrain ranking |

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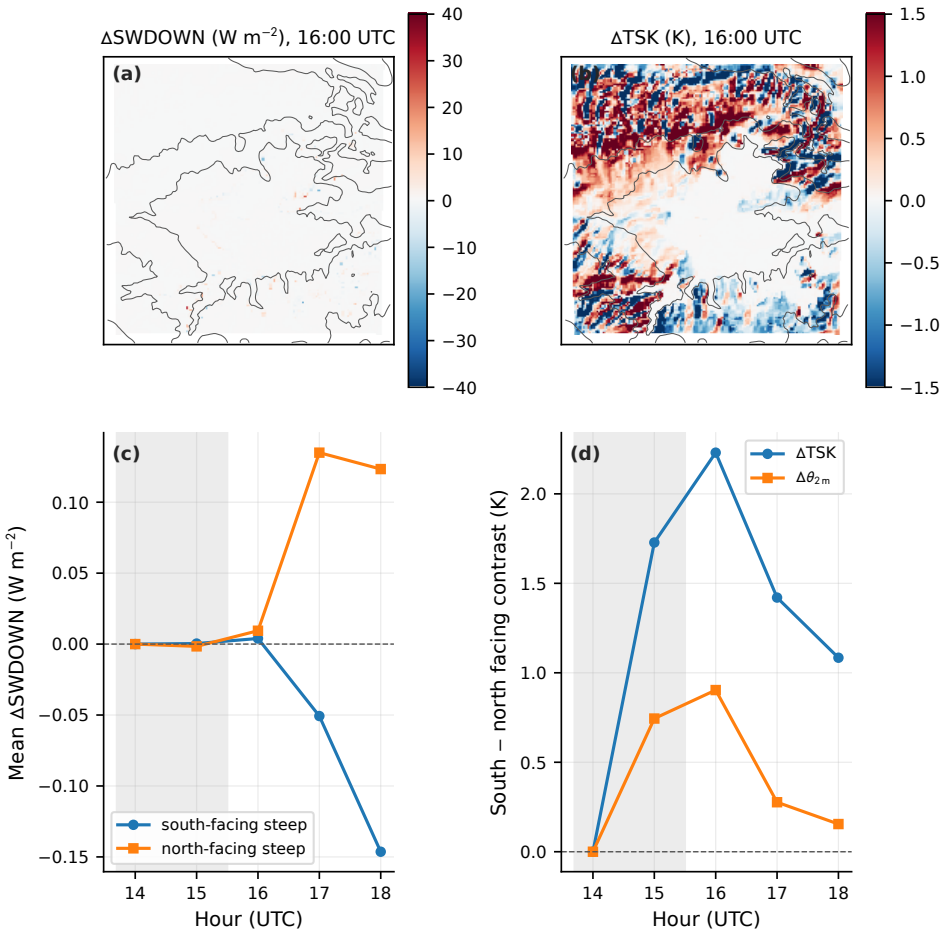



**Figure S1.** Diagnostic for the slope-radiation screen, calculated as X1 minus X0 on d02. The archived flat-plane shortwave field did not express the treatment, while skin temperature showed an aspect-organised response on steep terrain. The predeclared screen failed for this measurement-limited reason, and the figure does not support a configuration ranking or a claim that slope-aware radiation improved or degraded the cold pool.